

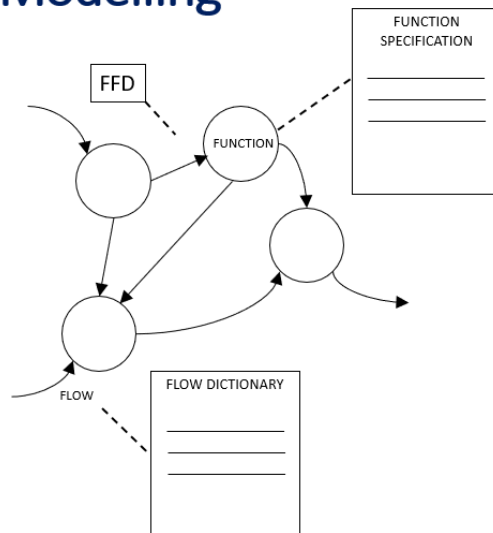
Tutorial Exercise 6

Functional & OO Analysis: Assessment project

1. Complete an FFD, or a series of hierarchical FFD diagrams for the system you are developing for your project.
2. Complete an object-oriented analysis of the system you are developing for your project. Think about the objects that are present. How are they related? How do they interact? What should they do and what properties should they have?
3. Design Domain Model based on the conceptual classes identified in the object-oriented analysis stage.
4. Have a general discussion with your colleagues about the approaches they are using. You could include a debate on FFDs and OO analysis, and whether they are appropriate tools to use, given the context of your project.

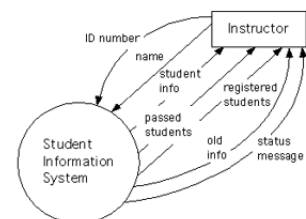
Functional Modelling

- A tool that highlights the **functionality** and the **inputs** and **outputs** to the identified **functions**
- Uses three sub-tools:
 - Function Flow Diagram (FFD)
 - Flow Dictionary
 - Function Specification



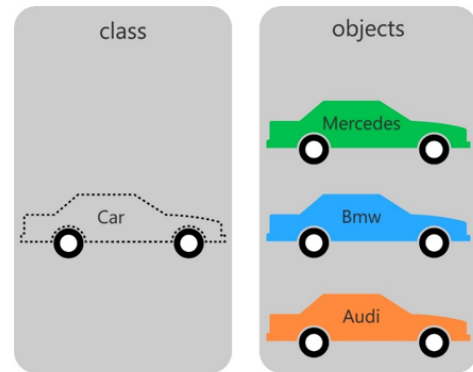
Context Diagram

- For large systems often a set of FFDs is required
- Can start with the **top level FFD**, which is referred to as the **Context Diagram**
- The **Context Diagram** shows the system in its **operational context** – the “super-system”. It has only one function
- It can be **decomposed into a number of functions that can be represented together on one diagram**. This is called **DIAGRAM 0**. Rule of thumb – no more than 7 functions in one diagram
- The functions in DIAGRAM 0 can be taken in turn and **decomposed further into lower-level FFDs**
- Can have a **hierarchy of FFDs**, each level providing lower level detail for the functions in the level directly above



Object-oriented Analysis

- The purpose, as with all other analysis methods, is to understand the application in terms of the system's functional requirements
- The difference is the means of expression
- Function / data analysis considers the system's data and behaviour separately
- Object-oriented analysis combines them and regards them as integrated objects
- Object-oriented analysis consists of the following five steps:
 1. Identifying the objects
 2. Organising the objects
 3. Describing how they interact
 4. Defining the operations on the objects
 5. Defining the objects internally



Domain Class Modelling

